# Pandas (Assignment Problems)

Let's do some practical exploration & analysis.

## Assignment Problems

**Q1**. Start by loading the **IRIS** dataset. [*Dataset Link:* Iris Flower Dataset]

**A.** Display the following information:

- First 10 rows
- Shape & Data types
- Summary statistics (mean, std, min, max)

**B.** Select only those rows where:

- *petal_length > 4.5*
- species = *"Iris-virginica"*

**C.** Group by *species* and compute:

- Average *sepal_length*
- Maximum *petal_width*
- Standard deviation of *sepal_width*

**D.** Create a new column: **"petal_ratio" = petal_length / petal_width**

- For each species, find the average petal_ratio.

**Q2**. Start by loading the **TITANIC** dataset. [*Dataset Link:* Titanic Dataset]

**A.** Display only columns: *Name, Sex, Age, Fare, Survived*

**B.** Select passengers who:

- Are female
- Fare > 30

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**C.** Group by *Pclass* and compute:

- Survival rate
- Average Fare
- Average Age

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